

CAPSTONE PROJECT PROPOSAL

Agricultural Supply Chain & Post-Harvest Loss Mitigation

Name	Role
Amos Ndungo	Data Engineering Lead : pipeline, synthetic data generation, ETL QA & Documentation Lead : testing, UAT notes, deployment readiness
Kevin Nyakundi	Modeling & ML Lead : hypothesis testing, feature engineering, modeling
Yvonne Wochuna	Dashboard & Business Analytics Lead - Visualization, stakeholder reporting, documentation, Storytelling

PROBLEM STATEMENT

Across Sub-Saharan Africa, smallholder horticultural supply chains experience significant post-harvest losses, with approximately 30–50% of fresh produce deteriorating before reaching wholesale markets. These losses reduce farmer incomes, increase food insecurity, and contribute to market price instability.

Current logistics operations rely on static transportation schedules and predetermined delivery routes, which fail because they do not adapt to changing environmental conditions such as ambient temperature, humidity, transport delays, road congestion, or fluctuating commodity prices.

We propose applying geospatial analytics, predictive machine learning, spatial-temporal feature engineering, and network optimization to estimate produce shelf-life degradation and dynamically determine optimal transport routes and market destinations. The project aims to reduce total transportation time by 15–25%, minimize post-harvest spoilage, and improve revenue retention for smallholder farmer cooperatives through data-driven logistics optimization.

DOMAIN AND CONTEXT

Industry: Agriculture Technology (AgTech), Supply Chain Analytics, and Smart Logistics.

The project focuses on improving the transportation of perishable horticultural produce from farms to regional wholesale markets within Sub-Saharan Africa.

Fresh produce such as tomatoes, vegetables, fruits, and leafy crops rapidly lose quality when exposed to prolonged transportation times, poor storage conditions, and adverse weather.

By combining agricultural production data, weather observations, logistics telemetry, and market pricing information, the project seeks to identify supply chain bottlenecks and recommend optimized transportation strategies that minimize spoilage while maximizing farmer profitability.

DATASET REQUIREMENTS & SCHEMA OVERVIEW

The analytical pipeline will integrate multiple publicly available datasets together with simulated logistics data.

Primary Agricultural Data

- FAOSTAT
- Regional agricultural production statistics
- Historical post-harvest loss indicators

Climate Data

- CHIRPS rainfall observations
- ERA5 climate datasets
- Daily temperature
- Relative humidity
- Rainfall intensity

Logistics Data

- Simulated transport telemetry
- Vehicle speeds
- Route distances
- Stopover durations
- Vehicle categories

Market Data

- Commodity market prices
- Wholesale demand
- Regional market locations

SUCCESS METRICS

The project will be considered successful if it achieves the following objectives:

- Predict shelf-life decay using regression models with strong performance measured by **RMSE, MAE, and R²**.
- Reduce transportation duration for perishable shipments by **15–25%**.
- Demonstrate measurable reductions in post-harvest crop spoilage.
- Increase projected revenue retention for participating farmer cooperatives.
- Identify major transportation bottlenecks across agricultural corridors.

INITIAL TIMELINE

Week	Focus	Deliverable
7	Data Collection & Cleaning	Clean agricultural, weather, and logistics datasets
8	Feature Engineering	Generate spatial-temporal variables, heat exposure indicators, route friction metrics
9	Machine Learning Modeling	Train Random Forest, XGBoost, and regression models
10	Route Optimization	Develop graph-based routing using NetworkX and Linear Programming
11	Dashboard Development	Build interactive Streamlit dashboard and visual analytics
12	Final Deployment	Complete documentation, presentation, and final demonstration

KEY RESEARCH QUESTIONS

1. Spoilage Prediction

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How accurately can transportation duration, road quality, temperature, and humidity predict post-harvest spoilage?

2. Supply Chain Bottlenecks

Which transportation corridors contribute most significantly to delays and crop deterioration?

3. Economic Impact

How much additional revenue can farmer cooperatives retain through optimized routing that reduces transportation time by 15–25%?

CORE PROJECT OBJECTIVES

- **Exploratory Data Analysis**

Analyze relationships between transportation variables, climate conditions, storage methods, and historical spoilage rates.

- **Predictive Modeling**

Develop machine learning models capable of estimating produce shelf-life degradation under varying transportation and environmental conditions.

- **Geospatial Optimization**

Design a graph-based routing system that recommends optimal market destinations based on transportation efficiency, commodity demand, and pricing.

- **Decision Support Dashboard**

Build an interactive dashboard that enables logistics managers to monitor transportation risks, visualize spoilage hotspots, and recommend optimal dispatch routes.

TECHNICAL APPROACH & METHODOLOGY

PHASE	ANALYTICAL FOCUS	METHODS & TOOLS	EXPECTED OUTPUTS
Data Ingestion	Data cleaning, spatial matching, timestamp alignment	Pandas, NumPy, GeoPandas	Cleaned master tabular & GIS dataset
Feature Eng.	Calculating thermal heat exposure during transit, route friction indices	Spatial Join, Haversine metrics, Feature Scaling	Engineered transit risk indicators
Modeling	Spoilage rate prediction & demand forecasting	Scikit-learn (Random Forest, XGBoost), Prophet	Evaluated ML models (RMSE, R², MAE)
Optimization	Shortest path & market price yield maximization	NetworkX, Linear Programming (PuLP)	Dynamic routing decision algorithm
Dashboard	Visualization	Streamlit, Plotly	Interactive decision-support system

EXPECTED CAPSTONE DELIVERABLES

Comprehensive Data Pipeline

A fully documented Python pipeline for data ingestion, preprocessing, feature engineering, and predictive modeling.

Analytical Report

A comprehensive evaluation of post-harvest loss drivers together with policy recommendations and logistics optimization strategies.

Interactive Dashboard

A Streamlit dashboard visualizing:

- Spoilage hotspots
- Transportation bottlenecks
- Weather conditions
- Market price opportunities
- Recommended transport routes

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